Son of Surya

GUIDE
R1
FUNDAMENTALS

FREE FOUNDATION GUIDE | BEGINNER

Electricity Basics for Solar

Volts, amps, watts, kW vs kWh, AC vs DC — the numbers on every bill, panel, inverter, battery and quotation, explained from zero for Indian homes and farms.

V	A	W	kWh	AC	Wp
VOLT	AMPERE	WATT	UNIT	/ DC	WATT-PEAK

“Understand your solar before you buy your solar.”

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R1 About This Guide

Guide	**R1 — Electricity Basics for Solar** (V, A, W, kW vs kWh, AC vs DC)
Category · Level · Price	Solar Fundamentals · Beginner (10th-standard friendly) · **Free**
For	Homeowners, farmers and shop owners considering solar; students starting out
Prerequisites	None — basic multiplication and division
Read next	R2 Your Electricity Bill · R3 Sun, Site & Shade · SOS-01 Solar Starter · R11 Solar Dictionary

The problem

Every solar conversation is full of numbers — 3 kW, 540 Wp, 1 kVA, 150 Ah, 400 units — and most buyers cannot tell what each one measures or whether two of them can even be compared.

The knowledge gap

The biggest confusion: **kW (power — how fast)** vs **kWh (energy — how much)**. Others: AC "tons", battery Ah, and believing grid-connected solar runs the house during power cuts.

The decision

After reading, you can tell which numbers on a bill, label or quotation matter, check that a quote's numbers are consistent, and make a first estimate of the system size you need.

The action

Collect 12 months of bills, read your appliance labels, work out your daily kWh, and ask every vendor: *"Is that kW or kWh — and how did you calculate it?"*

Each key idea is explained four ways: Simple Technical Practical Decision . Boxes marked ILLUSTRATIVE EXAMPLE are realistic but invented — not real customers or projects.

01 What You Will Learn

You will understand…	…so that you can
Volts, amps, ohms — push, flow and resistance	Read rating plates on panels, inverters, batteries and appliances
Power (W, kW) vs energy (kWh, "units")	Never confuse a 3 kW system with 3 units again
Energy = power × time	Calculate what each appliance costs you in units
AC vs DC, Wp, VA, power factor, Ah	Compare panels, inverters, home UPS units and batteries correctly
How energy flows in a grid-connected Indian home	Know why most solar systems switch off in a power cut
Indian regional solar output	Make a first, realistic estimate of system size

02 Why This Matters

A rooftop system lasts decades and every promise about it is written in electrical units. If you cannot read them, you cannot compare quotations or check a promise.

You hear...	What you need to know
"A 5 kW system will make your bill zero."	kW is size; your bill is in kWh. Size must come from your **consumption**. Fixed charges usually remain.
Quote A: "3 kW". Quote B: "3.24 kWp".	kWp is panel (DC) capacity; kW is often the inverter (AC). Different parts.
"Solar will give backup."	A standard grid-tied system **switches off** in a power cut. Backup needs a battery + hybrid inverter.
"150 Ah battery."	Ah is not energy. **Wh = V × Ah.**

03 Fundamentals — Starting From Zero

Electricity is the movement of tiny charged particles (electrons) through a conductor such as copper. Engineers often compare it to water in pipes.

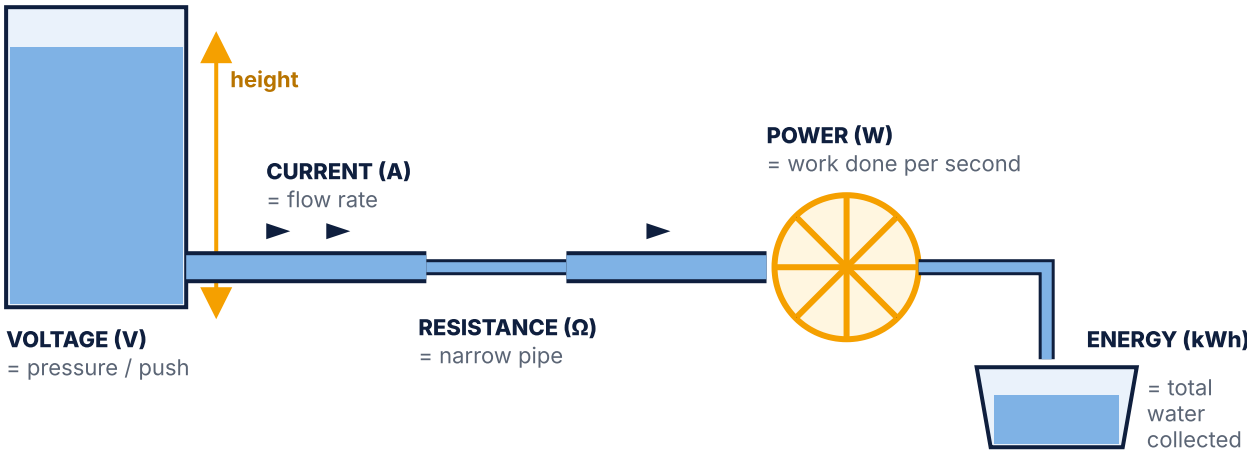


Figure 1. The water-pipe analogy. Voltage pushes, current flows, resistance restricts, power is the rate of work, and energy is the total work done over time.

Quantity (unit)	Simple	Technical	In your solar project	Decision use
Voltage (V)	How hard electricity is pushed	Potential difference; 1 V = 1 joule per coulomb	Socket ≈ 230 V AC; panel strings reach hundreds of V DC	Compatibility and danger

Current (A)	How much is flowing	Rate of charge flow; 1 A = 1 coulomb/second	Decides cable thickness, fuses, breakers	Ask for cable & breaker specs
Resistance (Ω)	A narrow pipe	Opposes flow; turns energy into heat	Long thin cables and loose joints waste energy	Good workmanship matters
Power (W, kW)	Speed of using energy	Energy per second; P = V × I (DC)	Appliance labels; system size (kW, kWp)	Tells you **size**, not units
Energy (kWh = unit)	Total used — distance, not speed	E = P × t; 1 kWh = 1,000 W for 1 h	Your bill; yearly solar generation; battery storage	**Money follows kWh**

The single most important idea: kW vs kWh



Figure 2. A car's speedometer shows how fast you are going (kW). Its odometer shows how far you have gone (kWh). Your electricity meter is an odometer.

SIMPLE

kW is how fast; kWh is how much. A 1,000 W heater for 3 hours and a 3,000 W heater for 1 hour both use 3 units.

TECHNICAL

Energy is power integrated over time — the area under the power curve. kWh = kW × hours.

PRACTICAL

A 3 kWp system gives roughly 3 kW at its best moment, but its **daily units** depend on hours of good sunlight — about 3–4 units per kWp per day across most of India (Section 06).

DECISION

Ask for generation and savings in **kWh per year**. Size quoted only in kW tells you nothing about your bill.

AC and DC

DC flows one way — panels and batteries make DC. **AC** reverses direction — India's grid runs at **50 Hz**, 50 cycles per second. The solar inverter converts DC to grid-synchronised AC.

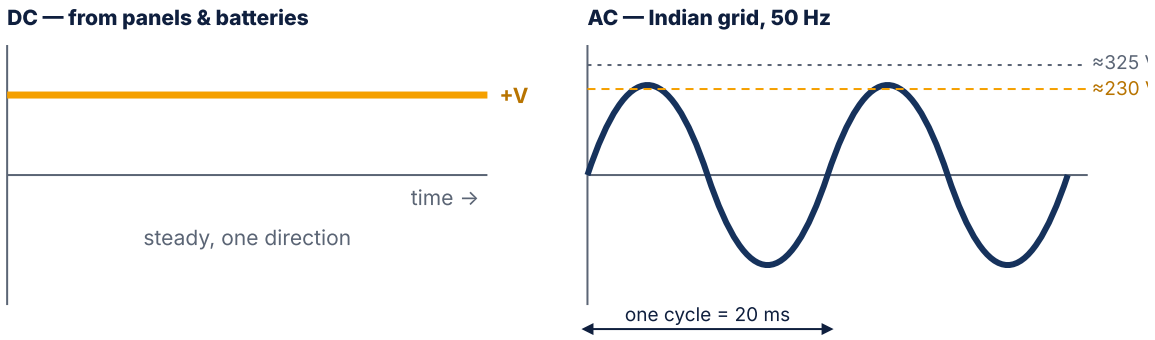


Figure 3. DC is steady. AC alternates. The "230 V" of an Indian socket is an RMS (effective) value; the waveform actually peaks at about 325 V.

04 Science & Engineering

Relationship	Formula	What it tells you
Ohm's law	$V = I \times R$	Push, flow and restriction are linked
Power (DC / AC)	$P = V \times I$ / $V \times I \times PF$	Same power at high V needs less current. AC power factor (0–1) reduces useful power
Energy	$E = P \times t$	Units on your bill = kW × hours
Wire heat loss	$P_{loss} = I^2 \times R$	Double the current → four times the heat
Battery energy	$Wh = V \times Ah$	Ah becomes energy only with voltage
AC peak	$V_{peak} = V_{RMS} \times \sqrt{2}$	230 V RMS peaks at about 325 V

WHY STRINGS RUN AT HIGH DC VOLTAGE

Loss depends on $I^2 \times R$. Raising voltage lowers current for the same power, so panels are wired in series strings of a few hundred volts (Calculation C7).

SINGLE- AND THREE-PHASE

Most homes: single-phase, nominal 230 V. Bigger loads, farms, shops: three-phase, ≈400 V between phases (230 × √3). Exact values come from your state supply code. **The inverter must match your connection.**

VA, KVA AND POWER FACTOR

Home UPS "inverters" are rated in VA. A 1,000 VA unit with PF 0.8 supplies at most ≈800 W. Grid-tied solar inverters are rated in kW. **Never compare VA with W directly.**

WATT-PEAK (WP) AND STC

A "540 Wp" panel is rated in a lab at 1,000 W/m², 25 °C cell temperature, AM 1.5 spectrum. Hot Indian roofs, dust, wiring and the inverter all reduce real output — so a 3 kWp array rarely delivers 3 kW AC. This is normal physics.

SAFETY CONSEQUENCE OF DC

Panels produce voltage whenever light falls on them — you cannot switch off the sun. DC does not pass through zero like AC, so DC arcs are harder to extinguish. **Rooftop DC work is for trained professionals only.**

05 How It Works in a Solar Home

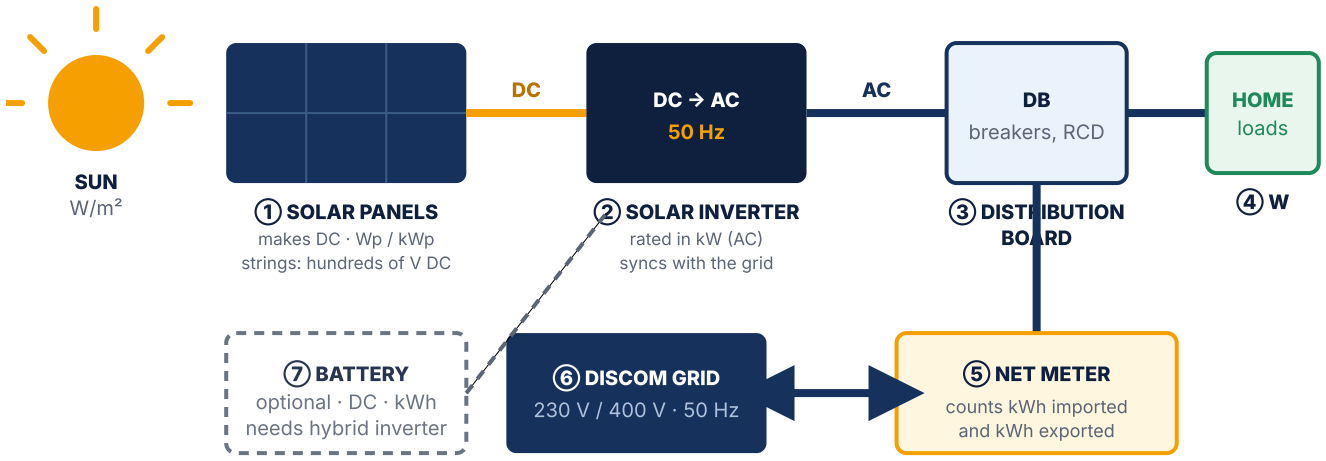


Figure 4. Energy flow in a grid-connected solar home, and the unit that belongs to each part. Simplified for learning — real installations also include isolators, surge protection, earthing and the metering arrangement your DISCOM specifies.

1. **Panels** turn sunlight (W/m²) into DC, rated in Wp / kWp.
2. **The inverter** converts DC to 50 Hz AC (rated in kW AC).
3. **Your appliances** use solar first; **surplus** flows to the grid through a bi-directional (net) meter, which counts exported kWh.
4. **Evenings and cloudy hours** you import from the grid; the bill settles imports and exports under your state's rules (SOS-08).

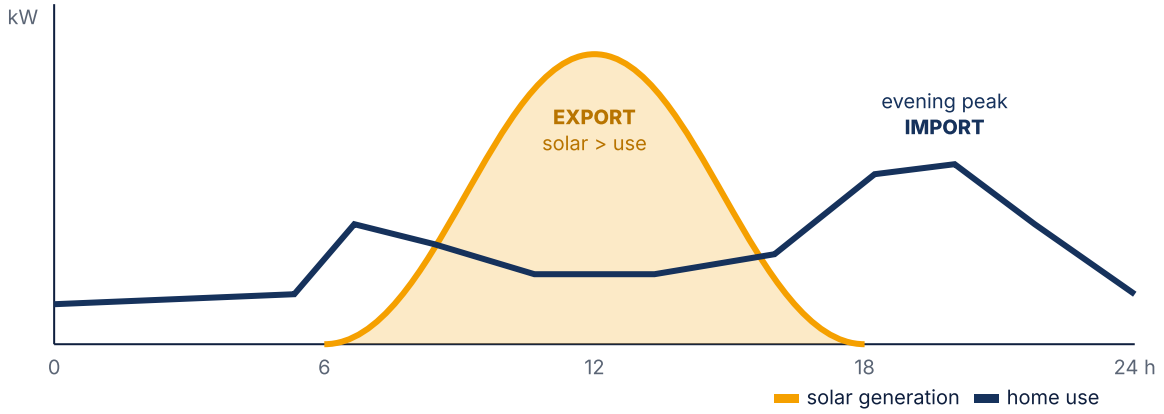


Figure 5. ILLUSTRATIVE EXAMPLE Typical shapes only. Solar peaks around midday; many homes use most electricity in the evening, when solar output is zero. The area under each curve is energy (kWh).

WHAT CHANGES IN INDIA? — THE POWER-CUT SURPRISE

Most rooftop systems are **grid-tied**. When grid power fails, the inverter **switches off automatically** (anti-islanding) so it cannot energise lines DISCOM staff may be repairing. Even at noon, no battery = no backup. Backup needs a **hybrid inverter + battery** (SOS-03, SOS-04).

Home UPS ("inverter")	Grid-tied solar inverter

Job	Store grid energy in a battery for power cuts	Convert solar DC to AC for home and grid
Rating	VA (e.g. 900 VA)	kW AC (e.g. 3 kW)
Power cut?	Works until battery runs low	Shuts down (unless hybrid with battery)
Cuts your bill?	No — it even loses some energy	Yes — every solar unit used is a unit not bought

YOU CAN SAFELY CHECK

- Labels and rating plates (without opening covers)
- Bill, meter and inverter display/app
- Daily and monthly kWh
- Warning lights, burning smell, heat — then call the installer

QUALIFIED PROFESSIONAL ONLY

- Any DC string, cable, connector or isolator
- Opening inverters, DBs, meters
- Batteries, earthing, surge protection
- Roof work and DISCOM grid connection

06 Indian Context

Factor	In India	Why it matters
Frequency	50.000 Hz reference; allowed band 49.900–50.050 Hz (CERC Grid Code 2023)	Equipment must be built for 50 Hz
Supply	Single-phase ≈230 V; three-phase ≈400 V; exact limits in state supply codes	Match inverter to phase; rural voltage swings can trip inverters
Bills	1 unit = 1 kWh; slab tariffs; fixed charges linked to sanctioned load (kW/kVA)	Savings depend on which slab solar removes
Sunlight	Most of India receives 4–7 kWh/m²/day (MNRE)	Good resource, but regional and seasonal
Heat, dust, monsoon	Hot roofs, dusty dry months, cloudy monsoon weeks	Real output below lab rating; plan on yearly totals
Farms	Pumps in HP (1 HP ≈ 0.746 kW); feeders often supply set hours	Convert HP to kW; think in pumping hours

Son of Surya regional planning figures

For first estimates, use the yearly-average figure for your region. These are deliberately conservative.

Region / state	Planning figure (units per kWp per day)	≈ units per kWp per year
Rajasthan	4.1	1,490
Gujarat	3.8	1,400

Andhra Pradesh · Telangana	3.7	1,360
Maharashtra · Madhya Pradesh	3.6	1,320
Karnataka · Tamil Nadu · Delhi NCR	3.4	1,235
West Bengal · Assam & North-East	2.9	1,050
Other states, hills, heavy-monsoon coasts	Use the Son of Surya calculator or a site tool	

HOW THESE FIGURES WERE MADE · VERIFY

Mid-points of published state yield benchmarks (1,225–1,750 kWh/kWp/yr, compiled from NREL, IRENA and NIWE data), reduced by 15% for real rooftop losses, ÷ 365, rounded. Cross-checks: MNRE's FAQ says 1 kWp makes *"4 to 5.5 units in a day"* on a **clear sunny day**; Tata Power-DDL's Delhi rooftop example implies ≈1,200 units/kWp/year (≈3.3/day). Well-designed, unshaded systems may beat these figures; shaded or dusty ones will not. Checked 7 Oct 2026.

UNITS INDIANS USE EVERY DAY

"1.5 ton AC"	Cooling capacity (1 ton ≈ 3.5 kW of heat removed) — **not** electricity used. Check the BEE label.
"1 HP pump"	≈ 0.746 kW mechanical output; electrical input is higher
"900 VA inverter"	× power factor (often 0.7–0.8; check plate) = watts
"150 Ah battery"	× battery voltage = Wh

GLOBAL PERSPECTIVE

Physics is identical everywhere; grids are not. Germany/EU: 230 V · 50 Hz (like India). USA: 120/240 V · 60 Hz. Saudi Arabia: 230 V · 60 Hz. UAE: 230 V · 50 Hz. Rules, tariffs and subsidies differ completely — never apply Indian rules abroad without checking.

07 Practical Application

Tool 1 — "Which number is this?" Look at the unit first; it tells you what kind of quantity you are reading.

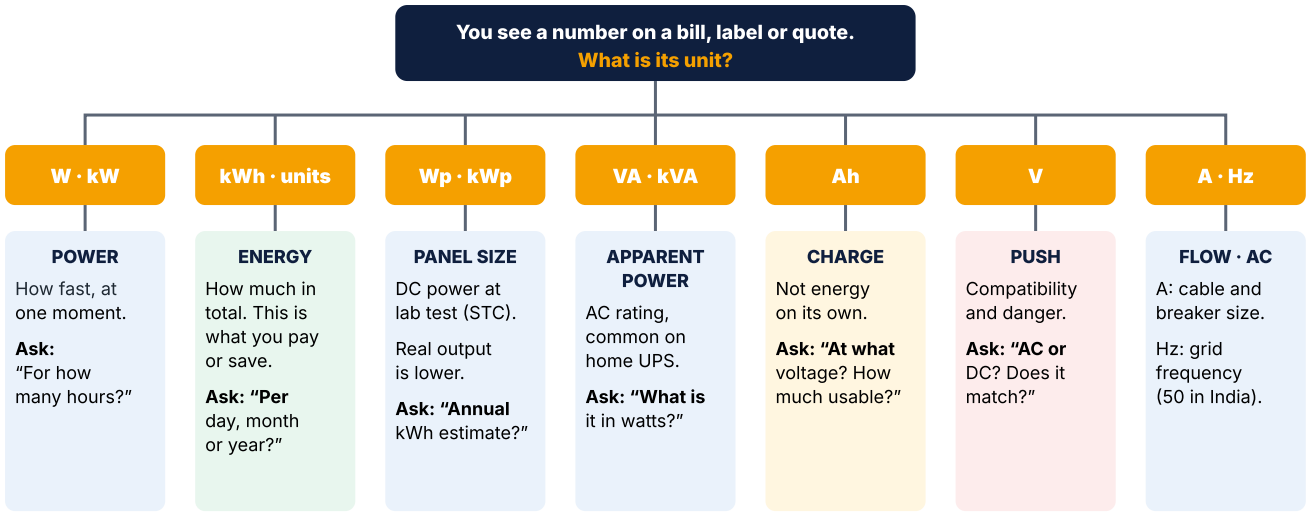


Figure 6. Decision tree: identify the quantity from its unit, then ask the right follow-up question.

Rating plate	Look for	It tells you
Appliance	V, Hz, W; BEE star label	Power when running; use BEE energy figure where given
Solar panel	Pmax (Wp), Voc, Isc	Pmax × number of panels = array kWp
Solar inverter	AC output (kW), max DC voltage, phase	Size and whether it matches your supply
Battery	Nominal V, Ah or kWh, chemistry	Energy = V × Ah; usable share by chemistry

Tool 2 — your bill shows both sanctioned load (**kW**, a power limit) and units consumed (**kWh**, energy). Guide R2 explains every line. **Tool 3 — after installation**, track "Today: kWh" and "Total: kWh" on the inverter app, and compare the same months year to year.

08 Calculations

LEARNING ESTIMATES — NOT A DESIGN

Real sizing needs site, shade, orientation, DISCOM rules and equipment checks (SOS-06 and a qualified designer).

C1 Energy used by one appliance

FORMULA	`E (kWh) = P (W) × t (h) ÷ 1,000`
ASSUMPTIONS	75 W ceiling fan, 12 h/day, 30 days
WORKING	75 × 12 = 900 Wh = 0.9 kWh/day → × 30 = 27 kWh
RESULT	**0.9 units/day · 27 units/month for one fan**
INTERPRETATION	Small loads running long hours add up. A ~30–35 W BLDC fan uses well under half.

C2 Monthly energy of a household

ASSUMPTIONS — **ILLUSTRATIVE EXAMPLE** Typical wattages and hours — use your own labels and habits.

Appliance	W	Qty	h/day	kWh/day
LED bulbs	9	6	6	0.324
Ceiling fans	75	3	12	2.700
Refrigerator (label: 250 kWh/year ÷ 365)	—	1	—	0.685
TV	100	1	4	0.400
Water pump (≈750 W input)	750	1	0.5	0.375
Mixer, washing machine, charging	—	—	—	0.775
Total without AC				**5.259**
AC, summer only (≈1,200 W average)	1,200	1	6	7.200

RESULT — **Non-summer ≈ 158 units/month (5.259 × 30) · Summer ≈ 374 units/month ((5.259 + 7.2) × 30)**

INTERPRETATION — More than double in summer — use **12 months of bills**. A fridge's compressor cycles, so use its label's yearly kWh, not W × 24.

C3 Current, and VA → watts

FORMULAS — `I = P ÷ (V × PF)` · `W = VA × PF`

WORKING — 3 kW inverter at 230 V, PF≈1: 3,000 ÷ 230 = **13.0 A**. 1,000 VA UPS, PF 0.8: 1,000 × 0.8 = **800 W** maximum load.

INTERPRETATION — Current sizes cables and breakers; VA is not watts.

C4 Battery backup time

FORMULA — `Runtime (h) = V × Ah × DoD × η ÷ Load (W)`

ASSUMPTIONS — 12 V 150 Ah lead-acid · 50% usable (conservative; follow the maker) · 85% UPS efficiency (assumed) · load 361 W (3 fans, 4 LEDs, TV)

WORKING — 1,800 Wh nominal → 900 Wh usable → 765 Wh delivered → 765 ÷ 361 = 2.12 h

RESULT — **≈ 2 h 7 min — not the 5 h that 1,800 ÷ 361 suggests**

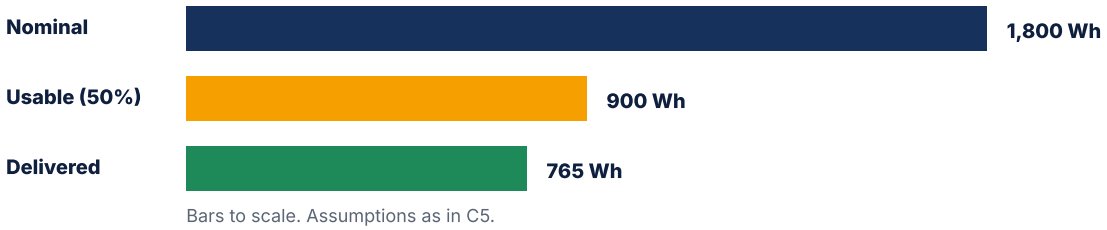


Figure 7. From label energy to energy your fans actually receive.

C5 Solar energy from system size (regional)

FORMULA	E_{year} = kWp × regional figure × 365
ASSUMPTIONS	3 kWp in Pune, Maharashtra → planning figure **3.6** units/kWp/day (Section 06)
WORKING	3 × 3.6 = 10.8 units/day → × 365 = 3,942 units/year → ÷ 12 ≈ 329 units/month
RESULT	**≈ 10.8 units/day · ≈ 3,940 units/year**
INTERPRETATION	Monsoon months fall below the average, dry sunny months rise above it.

C6 First estimate of size from your bills

FORMULA	kWp ≈ yearly units ÷ 365 ÷ regional figure
ASSUMPTIONS	C2 household in Pune: 8 months × 157.77 + 4 months × 373.77 units
WORKING	1,262.16 + 1,495.08 = 2,757.24 units/year → ÷ 365 = 7.55/day → ÷ 3.6 = 2.10 kWp
RESULT	**≈ 2.1 kWp — a starting point, not a final design**
INTERPRETATION	The same home in Rajasthan (4.1) needs ≈1.8 kWp; in West Bengal (2.9) ≈2.6 kWp. Location matters.

C7 Why high voltage cuts losses

WORKING	3,000 W through 0.1 Ω of cable. At 48 V: 62.5 A → 62.5^2 × 0.1 = **390.6 W lost (13%)**. At 400 V: 7.5 A → 7.5^2 × 0.1 = **5.6 W lost (0.19%)**.
INTERPRETATION	About 69× less loss — the reason panels are wired in series strings (and the reason DC work is for professionals).

09 Indian Case Study

ILLUSTRATIVE EXAMPLE — not a real customer or project

The Kulkarni family, Pune. Independent house, single-phase, sanctioned load 3 kW, a 900 VA home UPS. Bills: ≈160 units in winter, ≈370 in May. Two proposals arrive:

	Proposal A	Proposal B
Headline	"5 kW — your bill becomes zero!"	"2.16 kWp on-grid system"

Details	Price, “5 kW inverter”, panel brand	4 × 540 Wp; 2 kW single-phase inverter; ≈2,750–3,000 units/year with assumptions shown
Backup	Not mentioned	“No backup during power cuts” in writing

1. **12 bills** → ≈2,760 units/year → C6 → **≈2.1 kWp**. Proposal B (2.16 kWp) fits; A is more than double.
2. **A’s output:** 5 × 3.6 × 365 ≈ 6,570 units — ≈3,800 more than they use. Surplus credit depends on state rules and may be lower than the purchase tariff.
3. **“Zero bill”:** their bill has a fixed charge. The claim is unsupported.
4. **Sanctioned load 3 kW:** question for the DISCOM — what capacity is allowed, and would 5 kW need a load increase?
5. **Backup:** grid-tied solar will not replace their UPS in outages.

OUTCOME

They asked Proposal A for panel count × Wp, a yearly kWh estimate with assumptions, and the basis of the “zero bill” claim; they asked both to confirm DISCOM capacity rules. **The right size came from their kWh, not the biggest kW on offer.**

10 Cost & Economics

Money question	Driven by	Note
What will it cost?	kWp, inverter kW, battery kWh	Compare ₹ per kWp only when scope is identical
What will I save?	kWh used or credited × tariff	Slab tariffs: solar removes your costliest units first
What is surplus worth?	Exported kWh × settlement rate	State rules; may differ from your tariff (SOS-08)
What stays?	Fixed charges	Usually charged regardless of units

ILLUSTRATIVE EXAMPLE 240 units offset × assumed ₹8/unit = **₹1,920** that month. Change the tariff and the answer changes in proportion. Inverters usually have shorter warranties than panels; batteries lose capacity with age. Payback and loans: SOS-17, SOS-16. **No figure here is a guarantee.**

11 Common Mistakes

Mistake	Reality
“3 kW = 3 units a day.”	Daily units ≈ kWp × regional figure (≈ 9–12 for 3 kWp)
Sizing from one month’s bill	Use 12 months — consumption is seasonal

“Grid solar runs my home in a power cut.”	It shuts down; backup needs battery + hybrid inverter
Treating the home UPS as solar	A UPS stores grid power; it does not cut bills
Comparing batteries by Ah, or VA with W	Compare usable kWh; W = VA × PF
“1.5 ton AC uses 1.5 kW.”	Ton = cooling; check the BEE label
Expecting full Wp all day	Wp is a lab rating; heat, dust and losses reduce it
Planning on a sunny day’s output	Use yearly regional averages
Believing “zero bill” promises	Fixed charges and settlement rules remain
Thinking the AC breaker makes panels safe	DC is live in daylight — professionals only

12 Buyer Protection — Numbers to Demand

Every quotation should state	Unit	Why
Panel capacity, and Wp × quantity	kWp; e.g. 6 × 540 Wp	Verify the size yourself
Inverter rating and phase	kW AC; 1φ / 3φ	Must match your connection
Yearly generation estimate + assumptions	kWh/year	The number that drives savings
Battery (if any)	usable kWh, V, Ah, chemistry	Real backup
Backup capability	Yes/No, loads, hours	Avoids the power-cut surprise
Cables, protection, earthing; warranties	sizes, ratings, years	Safety and long-term cover

RED FLAGS — ASK QUESTIONS WHEN YOU SEE

Savings without kWh figures · “kW” without saying DC or AC · single-sunny-day generation promises · guaranteed zero bills or payback · battery in Ah without voltage · pressure to sign before comparing written numbers. A red flag is a reason to ask, not proof of bad intent.

13 Practical Checklist

BEFORE TALKING TO VENDORS

- [ ] 12 months of bills collected
- [ ] Yearly units: _____ kWh · daily average: _____
- [ ] Highest / lowest month: _____ / _____
- [ ] Phase: single / three · Sanctioned load: _____ kW
- [ ] Main loads listed from labels (W / HP / ton / VA)

WHEN A QUOTE ARRIVES

- [ ] kWp stated and Wp × quantity matches
- [ ] Inverter kW (AC) and phase stated
- [ ] Yearly kWh estimate with assumptions
- [ ] Estimate is reasonable for my region (C5)
- [ ] Battery in usable kWh; backup stated

- [ ] Future loads (EV, AC, pump): _____
- [ ] My region's planning figure: _____ → first estimate _____ kWp
- [ ] Backup needed? Which loads, how many hours?
- [ ] No guaranteed zero bill or payback
- [ ] Cables, protection, earthing, warranties listed

SAFETY

- [ ] I will not touch DC wiring, inverters, DBs or batteries
- [ ] I know whom to call for sparks, smells or alarms

SON OF SURYA FREE TOOL

Put your 12 bills and your region into the **Son of Surya Solar Calculator** at sonofsurya.com — it shows the size estimate **and every assumption behind it**, so you can compare it with any quotation.

14 Problem Solving

EASY **LED bulb**

9 W LED, 5 h/day, 30 days. How many units?

INTERMEDIATE **Lithium battery**

24 V 100 Ah LFP, 90% usable, 90% inverter efficiency, 400 W load. Runtime?

REAL-WORLD **Checking a quote in Ahmedabad**

A shop uses 3,300 units/year; sanctioned load 5 kW. Quote: 10 × 540 Wp panels, 5 kW inverter. (a) Array kWp? (b) Yearly units at Gujarat's figure (3.8)? (c) Size matching consumption? (d) Panels needed? (e) Questions to ask?

Complete solutions

Easy	9 × 5 = 45 Wh/day × 30 = 1,350 Wh = **1.35 units**. LEDs are cheap to run — target fans, ACs, pumps, geysers first.
Intermediate	24 × 100 = 2,400 Wh → × 0.9 = 2,160 → × 0.9 = 1,944 Wh → ÷ 400 = **4.86 h ≈ 4 h 52 min**. Fewer Ah than C4's battery (100 vs 150), yet more energy — Ah alone misleads.
Real-world	(a) 10 × 540 = **5.4 kWp** (inverter 5 kW AC is a different number). (b) 5.4 × 3.8 × 365 = **7,490 units/year**. (c) 3,300 ÷ 365 = 9.04/day ÷ 3.8 = **2.38 kWp**. (d) 2.38 ÷ 0.54 = 4.4 → **5 panels = 2.7 kWp**. The quote makes ≈2.3× the shop's use. (e) "Why 5.4 kWp? What yearly kWh and assumptions? How are exported units credited for my category? What does the DISCOM allow on 5 kW sanctioned load? Price for 2.7 kWp?"

15 FAQ

Q1 What is one "unit"?
1 kWh: 1,000 W for 1 hour, or 100 W for 10 hours.

Q2 Is 3 kW the same as 3 kWp?
Not necessarily — kWp is panels (DC), kW is often the inverter (AC). Ask for both.

Q3 How many units does 1 kWp make?
MNRE: 4–5.5 on a clear sunny day. Yearly averages are lower — see the regional table (2.9–4.1).

Q4 Why less than rated at noon?
Ratings are at 25 °C lab conditions; heat, dust and losses reduce output. Normal.

Q5 Does solar work in a power cut?
Grid-tied: no. Hybrid inverter + battery: yes.

Q6 Can I connect panels to my home UPS?
Only if the product is designed for it. Ask a qualified installer — never try it yourself.

Q7 Why do panels make DC?
Cells push electrons one way when lit. The inverter makes grid AC.

Q8 Is DC more dangerous?
Both can kill. Solar DC is live in daylight, high-voltage, and arcs persist. Professionals only.

Q9 Ah vs kWh?
Ah is charge; kWh is energy. Wh = V × Ah.

Q10 What does 50 Hz mean?
50 AC cycles per second — India's grid reference.

Q11 What is sanctioned load?
Your connection's approved maximum power. Many DISCOMs link allowed solar capacity to it — confirm locally.

Q12 Size to my highest summer bill?
Usually not; start from yearly units, then roof, budget and export rules.

Q13 Does bigger always save more?
Not once you exceed your use — surplus may be credited at a different rate.

Q14 Which inverter-app number to track?
kWh per day and month. kW shows only the moment.

Q15 My pump is 5 HP — how many kW?
≈3.7 kW output; input is higher. See SOS-12, SOS-13.

16 Glossary

AC — current that reverses direction; India 50 Hz.

Ampere (A) — unit of current.

Ah — battery charge; × V = Wh.

Anti-islanding — inverter shuts off when the grid fails.

BEE star label — efficiency label showing energy use.

DC — one-direction current from panels and batteries.

DISCOM — your electricity distribution company.

DoD — share of battery capacity used.

Earthing — safe path to ground for fault current.

Efficiency — useful output ÷ input.

Grid-tied — solar connected to the DISCOM grid.

Hybrid inverter — manages panels, battery and grid.

kW — 1,000 W; power.

kWh / unit — energy on your bill.

kWp — array DC rating at STC.

Net meter — records import and export.

Phase — single (homes) or three (larger loads).

Power factor — W ÷ VA, 0 to 1.

RCD / RCCB — trips on leakage current; protects people.

RMS — effective AC value (230 V).

Sanctioned load — approved maximum power of your connection.

Specific yield — units per kWp per day or year.

STC — 1,000 W/m², 25 °C, AM 1.5 test conditions.

String — panels in series.

VA / kVA — apparent power of AC equipment.

Volt (V) — electrical push.

Watt (W) — power; 1 joule per second.

Wp — panel rating at STC.
Full English–Hindi dictionary: Guide R11.

17 What To Do Next

R1 Basics ✓ → R2 Your Bill → R3 Sun & Site → SOS-01 Starter → Free Calculator

Have quotations already? Read R5 (Evaluating a Quotation) when it releases, and keep your documents in the free **Solar Project Passport** at sonofsurya.com.

18 Sources

Source	Used for	Checked
MNRE — Solar Overview (mnre.gov.in/en/solar-overview)	4–7 kWh/m²/day across most of India	7 Oct 2026
MNRE Rooftop Solar FAQ, via PIB (static.pib.gov.in, 28 Apr 2022)	4–5.5 units/kWp on a clear sunny day	7 Oct 2026
Tata Power-DDL, Details of Rooftop Solar (Dec 2020)	Delhi cross-check ≈1,200 units/kWp/yr	7 Oct 2026
Qbits Energy — state yield benchmarks (NREL/IRENA/NIWE compiled)	Basis of regional planning figures	7 Oct 2026
CERC Indian Electricity Grid Code 2023, Reg. 30(1) (via CERC Order 2/SM/2025)	50.000 Hz; 49.900–50.050 Hz	7 Oct 2026
CEA Safety & Electric Supply Regulations 2023 (cea.nic.in)	Safety context; RCDs in homes	summary only
IEC 60904-3 · BEE labelling · worldstandards.eu	STC · appliance labels · global voltages	reference

19 Disclaimer

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FREE FOUNDATION GUIDES

- **R1** Electricity Basics for Solar
- **R2** How to Read Your Electricity Bill
- **R3** Sun, Site & Shade
- **SOS-01** Solar Starter Guide
- **R11** Solar Dictionary (English–Hindi)

SPECIALISED GUIDES — RELEASING IN BATCHES

- **Home** — rooftop planning, sizing, net metering
- **Kisan** — farm solar, pumps, schemes
- **Business & institutions** — shops, schools, societies
- **Professional** — wiring, installation, EPC

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